

SILICATES

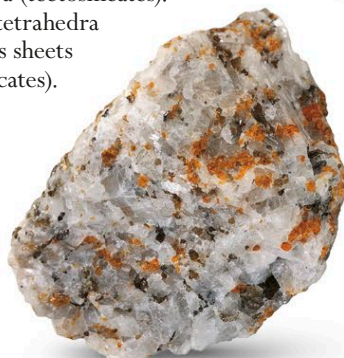
Silicates are the most common and largest group of minerals. The fundamental building blocks are tetrahedra of silicon and oxygen (SiO_4) together with other elements. They are subdivided into six groups based on the arrangement of the silica tetrahedra. Some form as isolated tetrahedra (nesosilicates), others occur in pairs (sorosilicates), and yet others have a three-dimensional network of tetrahedra (tectosilicates). Some silicates occur as chains of tetrahedra (inosilicates), while others form as sheets (phyllosilicates) or rings (cyclosilicates).

NESOSILICATES



HUMITE

Orthorhombic • H 6 • SG 3.2–3.32
A magnesium iron silicate fluorohydroxide, humite generally occurs as yellow to orange granular masses in metamorphosed limestones and dolomites.



NORBERGITE

Orthorhombic
H 6–6½ • SG 3.18

Norbergite mainly occurs as brownish yellow, white, or pink granular masses in metamorphic rocks. It is a magnesium silicate fluorohydroxide.



DATOLITE

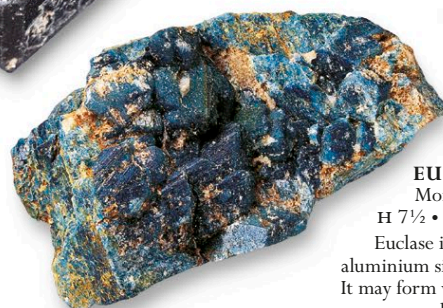
Monoclinic
H 5–5½ • SG 2.96–3

Datolite is a hydrous calcium boron silicate. Not very common, it is mainly found in veins or cavities in igneous rocks.



ANDRADITE

Cubic • H 6½–7 • SG 3.8–3.9
Yellowish green, brown, or black, andradite garnet is a calcium iron silicate. Cut gems are excellent at separating white light into colours.



EUCLASE

Monoclinic
H 7½ • SG 2.99–3.1

Euclase is a beryllium aluminium silicate hydroxide. It may form white, colourless, green, or blue prismatic, striated crystals.



MASSIVE DUMORTIERITE

DUMORTIERITE

Orthorhombic • H 7–8 • SG 3.21–3.41
Dumortierite is a silicate of aluminium, iron, and boron. It usually forms fibrous aggregates of radiating crystals, but it can also be massive.



KYANITE

Triclinic • H 5½–7 • SG 3.53–3.67
Kyanite is an aluminium silicate. Its bladed crystals in schist and gneiss formed at high pressures in the Earth.



PYROPE

Cubic • H 7–7½ • SG 3.58

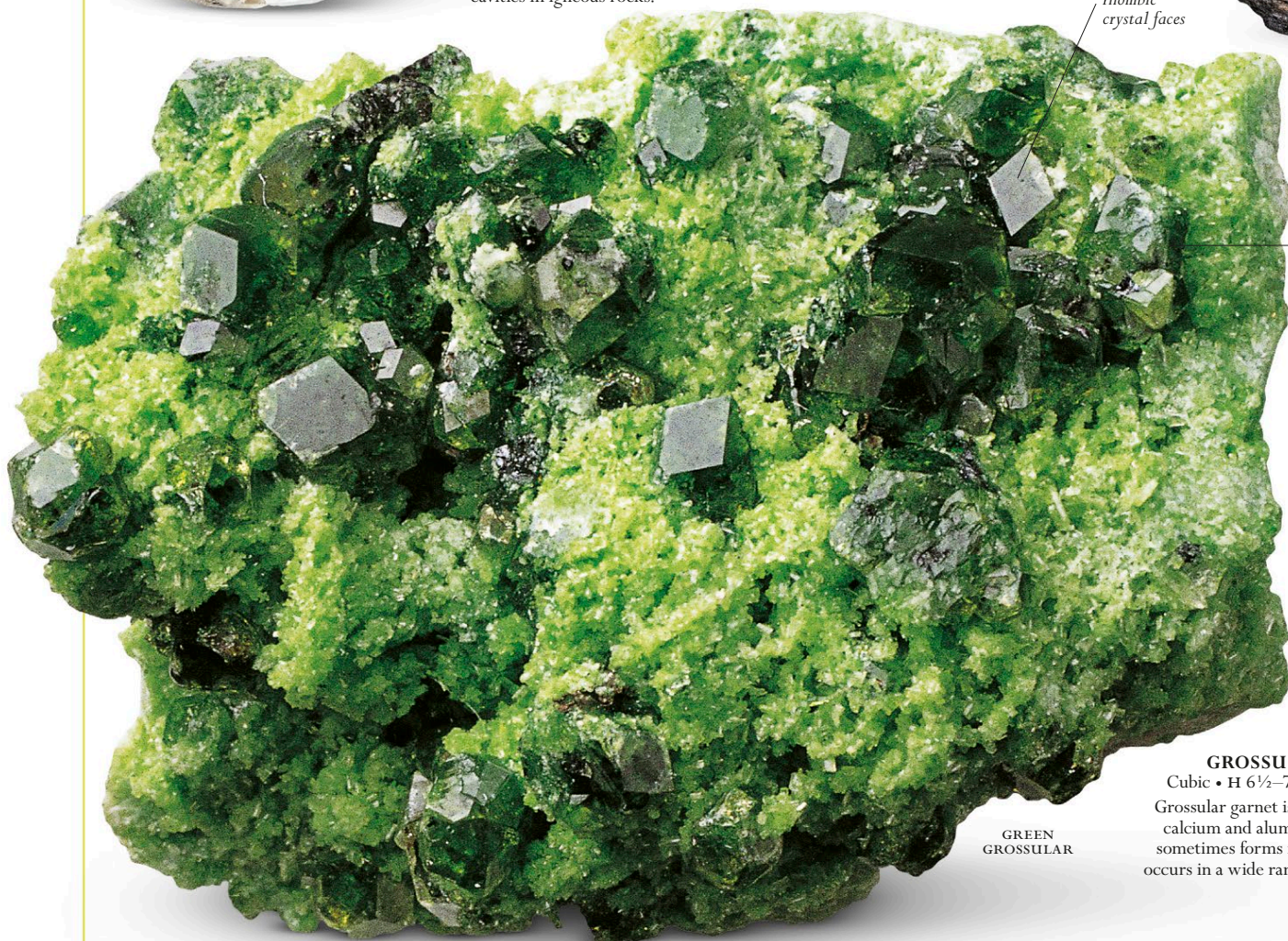
Pyrope garnet is a dark red magnesium aluminium silicate. It forms at high pressures in metamorphic and some igneous rocks.



ALMANDINE

Cubic • H 7–7½ • SG 4.32

The most common garnet, pinkish red almandine is an iron aluminium silicate. It is widely used as a gem.



GREEN GROSSULAR

GROSSULAR

Cubic • H 6½–7 • SG 3.59
Grossular garnet is a silicate of calcium and aluminium that sometimes forms in marble. It occurs in a wide range of colours.



RED GROSSULAR

rhombic
crystal faces

green colouring
due to vanadium

vitreous lustre

red colouring
due to iron